MyDesign Portfolio

Element K

Engineering Design Process Summary *(K1, K2)*

Here's a summarized breakdown of each step of the design process, organized by the specified elements:

1. Defining the Problem (Elements A–C)

- A. Identify the Need or Problem: Clearly state what issue needs solving or what opportunity exists. For us, the problem was that TV productions had unreliable methods to wield the light reflector board.

- B. Research and Gather Information: Collect relevant data, facts, and context to understand the problem better. To achieve this, our group interviewed Mr. O'Farrell and his students in order to achieve a better understanding of the problem.

- C. Define Constraints and Criteria: Establish the limitations (time, materials, cost) and requirements (functions, features) that the solution must meet. Going into this project as a team we knew that there wasn't a set time limit. There were a lot of questions that we had to answer such as how much money we would be spending and what materials we would use.

2. Ideation (Element D)

- D. Brainstorm and Generate Ideas: Creatively think of multiple possible solutions, encouraging quantity and diversity of ideas without judgment. In order to brainstorm, our group limited all distractions and went to the drawing board. We used sticky notes and a white board to log all of our ideas no matter how relevant they seemed.

3. Selection & Prototype Design (Elements D–F)

- D. Continue Brainstorming: Refine ideas from the ideation phase. After our ideation was complete we had many ideas in one place. So we grouped the ideas into categories based on the similar traits that each idea possessed.

- E. Evaluate and Select the Best Idea: Compare ideas against criteria and constraints to choose the most promising one. So, for the next design we had to take the best option from each category and compare them to each other, to get the overall winner.

- F. Develop Design Plans: Create detailed drawings or digital models outlining how the prototype will be built and function. So, after our group decided on an idea we created sketches with detailed dimensions, laying out how the mechanism would function.

4. Prototype Building (Element G)

- G. Construct the Prototype: Build a working model using appropriate tools and materials, following the design plans. After our sketches were complete our group went to Onshape to design and then 3D print the prototype.

5. Testing (Elements H–I)

- H. Test the Prototype: Try out the prototype under realistic conditions to see how well it performs. After the mechanism was printed we came up with a series of tests to test how practical and strong the clamp was.

- I. Record and Analyze Results: Collect data from the tests and determine how effectively the prototype meets the design criteria. We collected data from our tests such as the time it took to take the clamp on and off of the light pole.

6. Evaluation (Element J)

- J. Evaluate and Improve the Design: Assess strengths and weaknesses, then revise the design to enhance performance, efficiency, or user satisfaction. Obviously our first model wouldn't be the winner. For our group we redesigned our clamp four times to ensure we created the best possible product.

Lessons Learned About the Engineering Design Process *(K3)*

Element K: Reflection on the Engineering Design Process

Looking back on this project, there's a lot I've learned about what really makes the engineering design process work — and what can trip you up if you're not careful. These lessons go way beyond just this one design; they're things that any future engineer should keep in mind.

1. Start with a Deep Understanding of the Problem (A–C)
At first, it felt easy to jump right into sketching ideas, but we quickly realized how important it is to take time to clearly define the problem. Talking to users, doing background research, and outlining

constraints and criteria gave us direction — without that, we would've been designing blindly. In the future, I'll always spend more time up front making sure I fully understand what's actually needed before diving into solutions.

2. Don't Rush Through Brainstorming (D)
One thing that really stood out was how valuable it was to let all ideas surface — even the weird ones. Some of our best design elements came from unexpected combinations of early suggestions. Next time, I'd make sure the team creates a space where everyone feels comfortable throwing out ideas without worrying about being wrong. Creativity needs room to breathe.

3. Use a System to Pick the Best Idea (D–F)
We made the mistake early on of choosing a design because it "felt right" — not because it best met the criteria. That ended up costing us time later when the design didn't hold up. I learned the importance of using tools like a decision matrix to fairly evaluate each idea. It keeps things objective and makes sure we're not missing better options.

4. Build Fast, Learn Fast (G)
Trying to build a perfect prototype right away slowed us down. What worked better was making quick, simple versions of our idea so we could spot problems early. I now see that prototyping isn't about perfection — it's about learning fast and improving over time. Future projects should plan for multiple rounds of building and testing.

5. Test Like It's Real Life (H–I)
When we started testing, we didn't push the prototype hard enough or test it in realistic ways. That came back to bite us later when issues popped up that we hadn't seen coming. The big takeaway: don't just test to confirm your design works — test to break it. That's how you learn what really needs fixing.

6. Feedback and Reflection Are Where You Grow (J)
At first, it was frustrating when things didn't go as planned. But over time, we started seeing those moments as helpful feedback instead of failures. Reflecting as a team helped us improve our design — and our process. I now realize how important it is to regularly step back, ask what's working and what's not, and make honest adjustments.

Final Thoughts

The biggest lesson? The design process isn't just a checklist — it's a cycle. You go back, rethink, rebuild, retest. That's where real progress happens. If there's one thing I'd tell a future engineer, it's this: stay flexible, keep learning from every step, and don't be afraid to revise your work. That mindset is what leads to great design.